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A dynamic customer segmentation approach by combining LRFMS and multivariate time series clustering

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To successfully market to automotive parts customers in the Industrial Internet era, parts agents need to perform effective customer analysis and management. Dynamic customer segmentation is an effective analytical tool that helps parts agents identify different customer groups. RFM model and time series clustering algorithms are commonly used analytical methods in dynamic customer segmentation. The original RFM model suffers from the problems of R index randomness and ignoring customers' perceived value. For most existing studies on dynamic customer segmentation, time series clustering techniques largely focus on univariate clustering, with less research on multivariate clustering. To solve the above problems, this paper proposes a dynamic customer segmentation approach by combining LRFMS and multivariate time series clustering. Firstly, this method represents each customer behavior as a time series sequence of the Length, Recency, Frequency, Monetary and Satisfaction variables. And then, we apply a multi-dimensional time series clustering algorithm based on three distance measurement methods called DTW-D, SBD, and CID to carry out customer segmentation. Finally, an empirical study and comparative analyses are conducted using customer transaction data of parts agents to verify the effectiveness of the approach. Additionally, a detailed analysis of different customer groups is made, and corresponding marketing suggestions are provided.

Customer segmentation has become a research hotspot for customer relationship management in various industries. Customer segmentation can effectively assist companies in managing customer relationships and improving their services. It utilizes various advanced technologies such as statistical analysis, machine learning, pattern mining, and deep learning[1–3]. "Segmentation results are not static" has already attracted the attention of many scholars, and the underlying reasons are attributed to the changing time and space factors[4]. With the increase of customer information and the development of data analysis technology, the study of customer segmentation is no longer satisfied with the validity of a particular point in time[5]. And it focuses more on the behavioral trends of customers[6,7]. Therefore, the research on dynamic customer segmentation is significant.

The Recency, Frequency, and Monetary (RFM) model and time series clustering are commonly used analytical methods in dynamic customer segmentation. Their combination can provide accurate customer value analysis and better meet customer needs. However, these two methods also have some problems that need be solved. The traditional RFM model only considers the company's perceived value, neglecting the perceived value from the customer's perspective. In addition, the R-attribute in the RFM model has significant randomness. For dynamic customer segmentation, although most existing research focuses on the application of univariate time series clustering, there is increasing interest in the application of multivariate time series (MTS) clustering.

To solve these problems, this study proposes a dynamic customer segmentation approach by combining the Length, Recency, Frequency, Monetary and Satisfaction (LRFMS) model and MTS clustering. The method is called DCSLM (Dynamic Customer Segmentation by combining LRFMS and MTS). In the DCSLM, LRFMS attributes are extracted from customers' transaction data. Then, we utilize an MTS clustering approach that incorporates dynamic time warping based on multiple dimensions (DTW-D)[8,9], shape-based distance (SBD), and complexity-invariant dissimilarity (CID) to carry out customer segmentation. In MTS clustering, when only considering the overall matching of MTS, all variables are processed using DTW-D to generate a fuzzy

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membership matrix. When considering the overall matching of MTS and the contribution of each independent dimension, two composite fuzzy membership matrixes are generated based on DTW-D and SBD, as well as DTW-D and CID. For three fuzzy membership matrixes, clustering results can be obtained by applying spectral clustering (SP), affinity propagation (AP), and k-medoids clustering algorithms. The best-quality clustering results are obtained by cluster validity indices.

The rest of the paper is organized as follows. In "Related work" section, we provide a thorough description of related work. In "The proposed method" section, we describe the DCSLM. In "Empirical study" section, we verify the validity of the DCSLM using customer transaction data from parts agents. The results are discussed in "Analysis and comparisons of the DCSLM" section, and "Conclusions" section concludes the paper and discusses future research directions.

Related work

Dynamic customer segmentation is a customer behavior analysis method, which combines customer behavioral characteristics and attributes to divide customers into different groups, helping enterprises better understand customer needs[10–12]. Table 1 summarizes the research on customer segmentation in recent years.

Most scholars apply RFM to capture the dynamic behavior of customers to improve the effectiveness of dynamic customer segmentation. Syaputra et al.[4], Abbasimehr et al.[5,6,10,13,14], Sivaguru et al.[15], Liu et al.[16], and Yoseph et al.[17] have used RFM attributes and time series clustering to achieve dynamic customer segmentation. In addition to the original RFM, various modifications to the RFM model are used as segmentation variables[18]. Change of the RFM model is done by redefining one or more of the RFM variables or adding new variables, sometimes excluding some variables[18]. In the traditional retail industry, Huang et al.[19] modified the RFM model by adding the parameter C of social relations, providing more precise customer segmentation. Smaili et al.[20] proposed an RFM-D model, which added the diversity D of products purchased by a given customer, improving customer analysis and response prediction. Kao et al.[21], Wei et al.[22] and Husnah et al.[23] added the dimension of time length based on the RFM model to improve the accuracy of customer segmentation. Peker et al.[24] proposed a LRFMP model that adds length (L) and periodicity (P) based on the RFM model. The model performs well in classifying customers for the grocery retail industry and significantly improves the accuracy of customer segmentation. Although the above references improve the effectiveness of customer segmentation, there is a large randomness in the R index. In addition, the previous research on RFM only considers the firm's perceived value and ignores the customer's perceived value.

Clustering is most commonly used in RFM-based dynamic customer segmentation[25]. The clustering algorithms act as tools to deal with RFM attributes to achieve the customer segmentation objective[26]. Most of the studies in the literature have focused on two major properties of cluster analysis especially when dealing with time series data, including clustering algorithms and the similarity measure which is used to calculate the distance between the time series[27]. Some common clustering algorithms are k-means[7,10,19–21,24,28,29], k-medoids[10], fuzzy c-means (FCM)[10,15,17,23], DBSCAN[16,29], Spectral clustering (SP)[5], AP[29], and hierarchical clustering (HC)[5–7,13]. The second important property of the clustering methods is the similarity measure. Some common measures are SBD[5], CID[5,6,14], dynamic time warping (DTW)[4–6,13,14], temporal correlation coefficient (CORT)[5,6,13,14], correlation-based measure (COR)[13], discrete wavelet transform (DWT)[14], and Euclidean distance (ED)[6,13,14,28]. In the segmentation of pharmaceutical users in Indonesia, Syaputra et al.[4] modeled customer behavior as a time series of RFM and used DTW (CBRD) to achieve effective customer segmentation for the return product customers. Abbasimehr et al.[13] proposed a methodology for customer behavior analysis using time series clustering (CATC). The methodology modeled customer behavior as a time series of RFM, and used time series clustering including the similarity measure (i.e., ED, COR, DTW, and CORT) and HC, which validated its effectiveness and practicality by analyzing customer data from a bank. Zhou et al.[28] proposed a time series clustering algorithm based on smooth sequence ED measurement and k-means (TCBS), which analyzed user behavior patterns in online games, and effectively improved the accuracy and practicality of dynamic customer segmentation. Considering the time dimension of individual customer behavior, Abbasimehr et al.[5] proposed an analytical framework based on RFM model and time series clustering techniques (AFRC) for dynamic segmentation. This framework represented each customer behavior as a time series of RFM and used time series clustering algorithms for customer segmentation, which combined state-of-the-art similarity measures and some powerful time-series clustering algorithms to obtain the best quality clustering results, improving the refinement management and decision-making efficiency of enterprises. All the above methods effectively improve dynamic customer segmentation. However, most research has been developed primarily for univariate time series, and there has been a lack of effective solutions for dealing with MTS data.

In summary, it is challenging research to solve the problems of R index randomness and neglecting customer perceived value of traditional RFM model, and apply MTS clustering to dynamic customer segmentation. In this paper, we introduce the DCSLM method and validate its effectiveness using customer transaction data from parts agents.

The proposed method

In this section, we introduce the four main components of DCSLM, as shown in Fig. 1. The DCSLM consists of four main components: (1) representing customer behavior as a time series of LRFMS; (2) distance measures; (3) feature weighting and fuzzy membership matrix; and (4) clustering algorithms.

Representing customer behavior as time series of LRFMS

The RFM model is a crucial tool for measuring customer value and has found wide application in customer segmentation across various fields, including sales, marketing, and operations management[15]. However, the R index

Study	Contribution	Distance measures	Clustering algorithm	Modeling variables
Syaputra and Zulkarnain[4]	Proposing a method to identify similar groups of returned product customers via time series clustering.	DTW	–	RFM
Abbasimehr and Bahrini[5]	Proposing an analytical framework based on RFM model and time series clustering techniques for dynamic segmentation.	DTW, CID, CORT, SBD	HC, SP, k-shape	RFM
Abbasimehr and Shabani[6]	Developing an approach to identify similar groups of customers via time series clustering and then proposing an ensemble forecasting method to predict segment-wise behavior.	ED, DTW, CID, CORT	HC	RFM
Mosaddegh et al.[7]	Proposing a method for grouping customers based on CLV and tracking customer transition across segments using association rule mining.	–	k-means, HC	Variable utilized to quantify CLV
Abbasimehr and Baghery[10]	Proposing a time series clustering method with fine-tuned support vector regression for customer behavior analysis.	–	k-medoids, k-means, FCM, self-organizing map	RFM
Abbasimehr and Shabani[13]	Modelling customer behavior as a time series of RFM and using HC algorithm.	DTW, COR, CORT, ED	HC	RFM
Abbasimehr and Shabani[14]	Proposing a methodology for representing customer behavior with time series and then using distance-based time series clustering technique to identify customer groups.	ED, DTW, CID, CORT, DWT	–	RFM
Sivaguru[15]	Proposing a dynamic customer segmentation method using the modified dynamic fuzzy c-means clustering algorithm.	–	FCM	RFM
Liu and Chen[16]	Proposing a improved customer segmentation method based on RFM model and DBSCAN algorithm.	–	DBSCAN	RFM
Yoseph et al.[17]	Proposing a behavioral segmentation method to understand consumers in retail industry.	–	FCM, Expectation- maximization (EM)	RFM
Huang et al.[19]	Proposing a customer segmentation method based on RFMC model and k-means algorithm.	–	k-means	RFMC
Smaili and Hachimi[20]	Proposing a RFM-D model for improving customer analysis.	–	k-means	RFM-D
Kao et al.[21]	Applying LRFM model and k-means to evaluate an outfitter's customer values.	–	k-means	LRFM
Wei et al.[22]	Segmentation of patients using LRFM model and self-organizing maps in a children's dental clinic in Taiwan.	–	self-organizing maps	LRFM
Husnah et al.[23]	Applying the FCM algorithm and LRFM model to identify different groups of customers based on product category and brand category.	–	FCM	LRFM
Peker et al.[24]	Proposing a method based on LRFMP model and k-means algorithm to identify different customer segments in the grocery retail.	–	k-means	LRFMP
Zhou et al.[28]	Proposing a new smoothed sequence Euclidean distance metric and utilizing it for time series clustering of users in online games.	ED	k-means	Number of rounds and number of coins
Pavithra et al.[29]	Applying diverse different clustering techniques to identify different groups of customers in shopping centers.	–	K-means, AP, DBSCAN	Mall customers data

Table 1. Summary of past studies.

of the RFM model has a large randomness. For example, the R index of a new customer may be the same as a loyal customer's, resulting in an incorrect assessment of the customer's value. Additionally, the three-dimensional indexes of the original RFM model are selected from the perceived value of the firm. With the rapid development of the Internet, customers have become the source of a company's profit for its development. Therefore, customer satisfaction has become a critical factor in winning the market competition for companies.

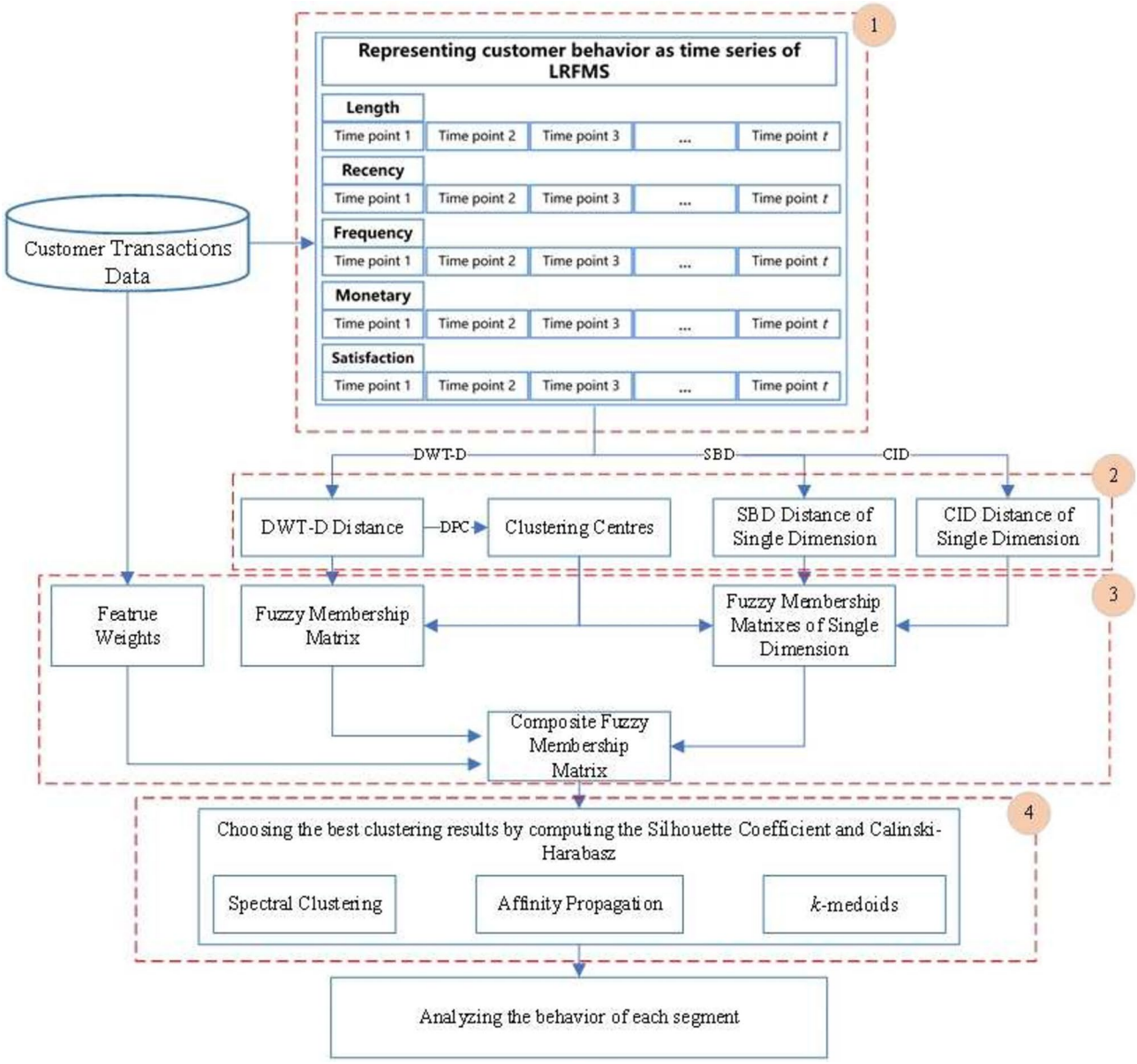


Figure 1. The overall schema of the proposed method.

To solve the above problem, we propose the LRFMS model. This model includes five indicators: Length (L_i), Recency (R_i'), Frequency (F_i), Monetary (M_i) and Satisfaction (S_i). L_i is the time elapsed between the first and last transaction in the ith interval, reflecting the customer's loyalty; R_i' denotes the average time elapsed between the end of the ith interval and the time of this customer's last p transactions in that interval; F_i represents the number of transactions during the interval; M_i represents the accumulated amount of customer transactions during the interval; and S_i represents customer satisfaction in that interval. Each indicator of the LRFMS is defined as follows:

$$L_i = last_{i1} - first_i, \tag{1}$$

where $last_{i1}$ represents the time of the customer's last transaction in ith interval and $first_i$ represents the time of the customer's first transaction in ith interval.

$$R_i' = \frac{1}{P}\sum_{b=1}^{P}(end_i - last_{il}), \tag{2}$$

where $end(i)$ represents the end time of the ith interval, and $last_{il}$ denotes the time of the customer's last lth transaction during ith interval. P represents the number of transactions closest to the end of the ith interval.

$$F_i = count(data), \tag{3}$$

where *count*(*data*) represents the number of transactions during *i*th interval.

$$M_i = sum(price), \tag{4}$$

where *sum*(*price*) is the accumulated amount of customer transactions during the interval.

$$S_i = \alpha \times CQ_i + \beta \times CD_i + \gamma \times CS_i, \tag{5}$$

where CQ_i represents the customer's satisfaction rating of product quality during *i*th interval; CD_i represents the customer's satisfaction rating with the timeliness of product delivery during *i*th interval; CS_i indicates the customer's satisfaction rating of after-sales service during *i*th interval; α, β, and γ are rating weights of customer satisfaction.

Distance measures

In this section, we introduce three state-of-the-art similarity measures, including DTW-D, CID, and SBD. They are used to calculate the distance between the time series.

Dynamic time warping

DTW, as a similarity measure, can find the optimal matching path between data in two arbitrary long time series by adjusting the correspondence between time points, which is robust to noise and can measure the similarity of time series more effectively[30]. Suppose there are two univariate time series $\tilde{X} = \{\tilde{x}_1, \ldots, \tilde{x}_i, \ldots, \tilde{x}_t\}$ and $\tilde{Y} = \{\tilde{y}_1, \ldots, \tilde{y}_j, \ldots, \tilde{y}_t\}$, where $\tilde{x}_i$ is a real number in the *i*th time point of $\tilde{X}$ and $\tilde{y}_j$ is a real number in the *j*th time point of $\tilde{Y}$. *t* is the length of a univariate time series. An alignment matrix of size $t \times t$ is constructed from $\tilde{X}$ and $\tilde{Y}$. Element (*i*,*j*) in the alignment matrix represents the *i*th element of $\tilde{X}$ corresponding to the *j*th element of $\tilde{Y}$[31]. The optimal alignment can be regarded as a warping path $M^{'}$ in the alignment matrix[31]. DTW can be defined as follows:

$$\mathrm{DTW}(\tilde{X}, \tilde{Y}) = \min \sum_{M'_{\grave{s}} \in M'} DS\left(M'_{\grave{s}}\right) \tag{6}$$

$$DS\left(M'_{\grave{s}}\right) = DS(i,j) = \left\|\tilde{x}_i - \tilde{y}_j\right\| \tag{7}$$

where $M' = \left\{M'_1, \ldots, M'_{\grave{s}}, \ldots, M'_{\grave{n}}\right\}$ represents a curved path, $\grave{n}$ is the total number of elements in the path. $M'_{\grave{s}} = (i,j) \in M'$ indicates the coordinate of the $\grave{s}$th point on the path, and $DS\left(M'_{\grave{s}}\right) = DS(i,j)$ represents the distance between $\tilde{x}_i$ and $\tilde{y}_j$. There are many paths like M'. All these paths form the path space M''. The goal is to obtain the optimal warping path that characterizes the path that minimizes the overall distance between $\tilde{X}$ and $\tilde{Y}$[10].

DTW uses dynamic programming to find estimated minimal warping path on an element-wise cost matrix given a cost function $\left\|\tilde{x}_i - \tilde{y}_j\right\|$[32]. To limit the warping path and prevent overfitting, we employ constraints. Specifically, we utilize the asymmetric slope constraint defined by the recurrence:

$$D(i,j) = DS(i,j) + DM(i,j) \tag{8}$$

$$DM(i,j) = \min[D(i-1,j), D(i,j-1), D(i-1,j-1)] \tag{9}$$

where $D(i, j)$ is the cumulative minimum cost at the *i*th and *j*th element, and $DM(i, j)$ is the minimum of the three elements' cumulative distance values. The last element $D(t, t)$ of D is DTW distance between the time series $\tilde{X}$ and $\tilde{Y}$.

DTW is commonly used to measure the similarity between two univariate time series objects. However, it is not suitable for calculating the similarity between two MTS objects. In this study, we use DTW-D, which takes into account the overall matching of MTS[33,34]. A z-dimensional MTS with a length of *t* is represented as follows.

$$X = [\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_\mathrm{t}]^T = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1z} \\ x_{21} & x_{22} & \cdots & x_{2z} \\ \vdots & \vdots & \vdots & \vdots \\ x_{t1} & x_{t2} & \cdots & x_{tz} \end{bmatrix} \tag{10}$$

where $\mathrm{x_i}$ represents a set of elements at the *i*th time point[31]. Suppose there are two MTS X and Y. DTW-D returns an optimal alignment utilizing Eq. (11).

$$\mathrm{dt}\left(\mathrm{x_i}, \mathrm{y_j}\right) = \sqrt{\sum_{m=1}^{z} \left(x_{im} - y_{jm}\right)^2} \tag{11}$$

where $\mathrm{dt}\left(\mathrm{x_i}, \mathrm{y_j}\right)$ represents the multivariate distance between $\mathrm{x_i}$ and $\mathrm{y_j}$, $\mathrm{x_i}$ is the *i*th time point of X, and $\mathrm{Y_j}$ is the *j*th time point of Y.

CID

Proposed by Batista et al.[30], it uses the complexity difference between two-time series as a correction factor for the existing distance metric[5,30]. A general CID measure is defined as:

$$\mathrm{CID}(\tilde{X}, \tilde{Y}) = F(\tilde{X}, \tilde{Y}) \cdot OD(\tilde{X}, \tilde{Y}) \tag{12}$$

where $OD(\tilde{X}, \tilde{Y})$ represents the distance metric (e.g., Euclidean distance), and $F(\tilde{X}, \tilde{Y})$ depicts the complex correction factor by Eq. (13):

$$F(\tilde{X}, \tilde{Y}) = \frac{\mathrm{MAX}\{CE(\tilde{X}), CE(\tilde{Y})\}}{\mathrm{MIN}\{CE(\tilde{X}), CE(\tilde{Y})\}} \tag{13}$$

where $CE(\tilde{X})$ and $CE(\tilde{Y})$ are the complexity estimators. For time series $\tilde{Y}$, $CE(\tilde{Y})$ is defined as:

$$CE(\tilde{Y}) = \sqrt{\sum_{j=1}^{t-1} \left(\tilde{y}_j - \tilde{y}_{j+1}\right)^2} \tag{14}$$

Shape-based distance measurement

SBD is a novel measurement approach that handles distortions in amplitude and phase accurately, by efficiently using cross-correlation through the normalization of a shape-based distance measure[34]. It is defined as:

$$\mathrm{SBD}(\tilde{X}, \tilde{Y}) = 1 - \max_{\rho} NCC_{\rho}(\tilde{X}, \tilde{Y}) \tag{15}$$

where NCC represents the normalized cross-correlation measure, which is calculated by applying the Fast Fourier Transformation algorithm, and ρ is the position that $NCC_w(\tilde{X}, \tilde{Y})$ is maximized.

Feature weight and fuzzy membership matrix

Feature weight

In most datasets, each feature has a different value, and some features contain more information than others. Therefore, when clustering, it is essential to compute a feature weight vector[31]. Li et al.[31] used range to compute the feature weights. The range is affected by the extremum and does not truly reflect the dispersion of the data. Therefore, this study uses the variance to calculate the feature weights.

The dataset to be clustered with n MTS objects is called $M = \{X_1, X_2, \ldots, X_n\}$.$W = \{w_1, w_2, \ldots, w_z\}$ represents the feature vector of an MTS dataset, and w_l can be computed based on the M as follows:

$$w_l = \frac{\sum_{\check{i}=1}^{n} \left| X_{\check{i}}^{l} - \overline{X^l} \right|}{\sum_{a=1}^{z} \left(\sum_{\check{i}=1}^{n} \left| X_{\check{i}}^{a} - \overline{X^a} \right| \right)}, l \in \{1, 2, \ldots, z\} \tag{16}$$

where $X_{\check{i}}^{l}$ denotes the lth dimension of the $\check{i}$th MTS object in M. $X_{\check{i}}^{a}$ represents the ath dimension of the $\check{i}$th MTS object in M. $\overline{X^l}$ represents the average value among all univariate time series in the lth dimension of M. $\overline{X^a}$ represents the average value among all univariate time series in the ath dimension of M.

Center choosing by fast searching and finding density peaks

Rodriguez and Laio[35] proposed a clustering by fast search and find of density peaks (DPC), which automatically finds the cluster centers and is suitable for arbitrary-shaped datasets. The main idea of the DPC algorithm is that the cluster center exhibits a higher local density in the corresponding clusters and is positioned at a relatively larger distance from other cluster centers compared to its local data points[31]. Compared to traditional clustering algorithms, it is simpler and more efficient[35]. The DPC algorithm consists of two important parameters for each data point $\check{i}$: its local density $\rho_{\check{i}}$, and the distance to the nearest high-density point $\delta_{\check{i}}$. For each point $\check{i}$, its local density $\rho_{\check{i}}$ is defined as follows:

$$\rho_{\check{i}} = \sum_{\check{i} \neq \check{j}} \chi\left(d_{\tilde{ij}} - d_c\right) \tag{17}$$

where if we assume $\check{x} = d_{\tilde{ij}} - d_c$, then $\chi(\check{x}) = \begin{cases} 1 & \check{x} < 0 \\ 0 & \check{x} \geq 0 \end{cases}$. $d_{\tilde{ij}}$ denotes the Euclidean distance between $\check{i}$ and $\check{j}$, and d_c denotes the cutoff distance. d_c is 1% to 2% of the total number of points in the dataset. The distance to the nearest high-density point $\delta_{\check{i}}$ is computed as follows:

$$\delta_{\check{i}} = \min_{\check{j}:\rho_{\check{j}} > \rho_{\check{i}}} \left(d_{\tilde{ij}}\right) \tag{18}$$

In this study, we chose the cluster centers according to a decision graph, and then select points with large $\rho_{\check{i}}$ and $\delta_{\check{i}}$ values as cluster centers. The following are the detailed steps of the DPC algorithm[31].

Input: Distance matrix of dataset D, cutoff distance d_c
Output: Cluster centers $\check{C} = \{\check{c}_1, \check{c}_2, \ldots, \check{c}_k\}$

for each $\check{i} \in [1, n]$ **do**
  Calculating local density $\rho_{\check{i}}$ according to Eq. (17)
end for
for each $\check{i} \in [1, n]$ **do**
  for all $\check{j}$ such that $\rho_{\check{j}} > \rho_{\check{i}}$ **do**
    Calculating distance $\delta_{\check{i}}$ according to Eq. (18)
  end for
end for
Plot the decision graph with ρ as the horizontal axis and δ as the vertical axis
Mark the point where both ρ and δ are relatively high as the cluster center

Algorithm 1. Center choosing based on DPC

Fuzzy membership matrix
The fuzzy membership matrix measures the relevance between two objects, helping to optimize clustering algorithms and produce more accurate results. It consists of a series of elements called fuzzy members. Each element represents the degree of belonging of a given cluster. It takes values ranging from zero to one. Converting original MTS objects into a fuzzy membership matrix F is computed as follows:

$$f_{\check{i}\check{j}} = \frac{\left(\frac{1}{\text{distance}\left(X_{\check{i}}, X_{\check{C}_{\check{j}}}\right)}\right)^{\frac{1}{2}}}{\sum_{S=1}^{k}\left(\frac{1}{\text{distance}\left(X_{\check{i}}, X_{\check{C}_S}\right)}\right)^{\frac{1}{2}}} \tag{19}$$

where distance represents the distance measurement. Cluster centers are $\check{C} = \{\check{c}_1, \check{c}_2, \ldots, \check{c}_k\}$, where k is the number of clusters. $\check{C}_s$ represents the index of the sth cluster center. $\check{C}_{\check{j}}$ represents the index of the $\check{j}$th cluster center.

Composite fuzzy membership matrix
In this section, we use the methodology proposed in Li and Wei[31] to generate the compound fuzzy membership matrixes $O_{DTW-D+CID}$ and $O_{DTW-D+SBD}$. The complete process of calculating the fuzzy membership matrix is shown in Algorithm 2.

Input: SBD distance matrices $D_{sbd} = \{D^1_{sbd}, D^2_{sbd}, \ldots, D^z_{sbd}\}$, CID distance matrices $D_{cid} = \{D^1_{cid}, D^2_{cid}, \ldots, D^z_{cid}\}$, DTW distance D_{dtw}, feature weights $W = \{w_1, w_2, \ldots, w_z\}$, $Distance_options = \{d_{DTW-D}, d_{DTW-D+CID}, d_{DTW-D+SBD}\}$
Output: Fuzzy membership matrix O

Generate the cluster centers $\check{C} = \{\check{c}_1, \check{c}_2, \ldots, \check{c}_k\}$ according to Algorithm 1
Calculate the fuzzy membership matrix F_{dtw} based on DTW-D distance according to Eq. (19)
if option is d_{DTW-D} **then**
  Fuzzy membership matrix $O = F_{dtw}$
else if option is $d_{DTW-D+SBD}$ **then**
  Calculate the fuzzy membership matrixes $F_{sbd} = \{F^1_{sbd}, F^2_{sbd}, \ldots, F^z_{sbd}\}$ for each dimension based on SBD distance according to Eq. (19)
  Combine Z fuzzy membership matrices into one fuzzy membership matrix $F_s = \sum_{l=1}^{Z} w_l \cdot F^l_{sbd}$
  Composite fuzzy membership matrix $O_{DTW-D+SBD} = F_{dtw} + \frac{F_S}{z}$
  $O = O_{DTW-D+SBD}$
else
  Calculate the fuzzy membership matrix $F_{cid} = \{F^1_{cid}, F^2_{cid}, \ldots, F^Z_{cid}\}$ for each dimension based on CID distance according to Eq. (19)
  Combine Z fuzzy membership matrices into one fuzzy membership $F_c = \sum_{l=1}^{Z} w_l \cdot F^l_{cid}$
  Composite fuzzy membership matrix $O_{DTW-D+CID} = F_{dtw} + \frac{F_C}{z}$
  $O = O_{DTW-D+CID}$
end if

Algorithm 2. Fuzzy membership matrix

Time series clustering

AP[36], SP[37] and k-medoids[10] are the three most popular clustering algorithms[5]. AP and SP have many advantages, such as being suitable for high-dimensional data and automatically determining the number of clusters. These advantages make these two methods widely applicable and essential in analyzing and applying multi-dimensional time series data. The k-medoids algorithm is a method suitable for time series clustering, which has good robustness to noise and outliers, and can handle large datasets. In practical applications, different clustering algorithms can be selected for time series clustering according to their needs to achieve better clustering results[5]. Since AP and k-medoids are widely used, we only introduce the spectral clustering.

Spectral clustering is a graph-theoretic algorithm used for data classification. It treats each data point as a node in a graph and connects them based on their similarity. The graph is then partitioned to minimize the sum of weights between different subgraphs while maximizing the sum of weights within each subgraph. The following is a description of spectral clustering.

Input: Fuzzy membership matrix $O = \{O_1, O_2, \ldots, O_n\}$, number k of clusters

Output: Clustering results $C = \{C_1, C_2, \ldots, C_k\}$

Step 1: Transform the fuzzy membership matrix O into a similarity matrix S as follows.

$$s_{\tilde{ij}} = e^{\frac{-\left\|O_{\tilde{i}} - O_{\tilde{j}}\right\|^2}{2\sigma^2}} \tag{20}$$

where σ is the bandwidth value.

Step 2: Transform the similarity matrix S into the Laplacian matrix L as follows.

$$L = D^{-0.5} S D^{-0.5} \tag{21}$$

where D is a degree matrix, and its diagonal elements are the sum of node degrees.

Step 3: Perform eigenvalue decomposition on the Laplacian matrix L to obtain the first k eigenvectors $v_1, v_2, \ldots, v_k$ and corresponding eigenvalues, and create the eigenvector matrix $V = \left(v_{\tilde{ij}}\right)_{n \times k}$.

Step 4: Obtain matrix U by normalizing V using Eq. (22):

$$u_{\tilde{ij}} = \frac{v_{\tilde{ij}}}{\sqrt{\sum_{\tilde{j}} v_{\tilde{ij}}^2}} \tag{22}$$

Step 5: Each row of the matrix U is regarded as a data point, and these data points are clustered using the k-means clustering algorithm to obtain the clustering result C.

The time complexity of traditional spectral clustering is $O\left(n^3\right)$. Therefore, this study adopts the optimised spectral clustering method proposed in literature[38], which can effectively reduce the time complexity. For more information about this method, please refer to the relevant literature[38].

Each step of the DCSLM

The DCSLM takes the time-stamped customer transaction information as input and transforms the customer behavior into multi-dimensional time series data M for LRFMS. Then, some powerful clustering algorithms and state-of-the-art distance measures are combined to obtain well-separated and dense clusters[5]. Finally, based on the clustering results, marketing strategies are designed to meet customer needs and improve customer satisfaction. The main steps of the DCSLM method (MTS clustering) are as follows in Algorithm 3.

Input: MTS dataset $M = \{X_1, X_2, \ldots, X_n\}$, cluster number k, $Methods = \{AP, SP, k\text{-medoids }\}$, $Distance_options = \{d_{DTW-D}, d_{DTW-D+CID}, d_{DTW-D+SBD}\}$

Output: Cluster result $C = (C_1, C_2, \ldots, C_k)$

Calculate feature weights $W = \{w_1, w_2, \ldots, w_z\}$ according to Eq. (16)
Calculate multivariate DTW distance D_{dtw} according to DTW-D
Calculate the SBD distance matrices within each dimension $D_{sbd} = \left\{D_{sbd}^1, D_{sbd}^2, \ldots, D_{sbd}^z\right\}$ according to Eq. (15)
Calculate the CID distance matrices within each dimension $D_{\text{cid}} = \left\{D_{\text{cid}}^1, D_{\text{cid}}^2, \ldots, D_{\text{cid}}^z\right\}$ according to Eq. (12)
for each $option \in Distance_options$ **do**
    Generate fuzzy membership matrix O according to Algorithm 2
    for each $algorithm \in Methods$ **do**
        $(C_1, C_2, \ldots, C_k)$=$do_clustering(algorithm, O, k)$
    end for
    Select the best clustering model by computing cluster validity indices
    Analyze the behavior patterns of each customer segment over time and develop targeted marketing strategies
end for

Algorithm 3. The Proposed Method (DCSLM)

Empirical study

To implement the proposed method DCSLM, we use Python 3.9 programming language in Anaconda software version 2021. We obtain 300,000 customer transaction data of parts agents from the ASP/SaaS-based manufacturing industry value chain collaboration platform for 2019-2021[39,40]. We extract features such as customer ID, part name, part brand, quantity, and timing of purchase from the customer transaction data to form two customer transaction datasets[39]. The first dataset, consisting of "Engine Parts", "Clutch and Transmission Parts", and "Hydraulic Lift Parts" purchased by the customer, is called PD1. The second dataset, consisting of "Body and Interior/Exterior Parts" purchased by the customer, is called PD2. The third dataset, consisting of "Electric Parts" purchased by the customer, is called PD3.

Representing customer behavior as time series of LRFMS

In this step, LRFMS analysis generates time series data that capture changes in customer behavior over time. We first use min-max normalization to normalize the customer transaction data[41]. Next, the transaction data is aggregated into bins based on a two-week time interval. Finally, by considering the definition of LRFMS variables, time series of L, R′, F, M, and S are extracted every two weeks. Since we consider CQ_i, CD_i and CS_i to be equally important in customer satisfaction S, α, β, and γ are set to 1/3.

Multivariate time series clustering

In this step, first, the DPC algorithm is applied to directly generate k cluster centers. Then, three distance measures, namely, DTW-D, (DTW-D+ CID), (DTW-D+ SBD), are used to generate three fuzzy membership matrixes according to Algorithm 2. For three fuzzy membership matrixes, clustering results can be obtained by applying three clustering algorithms (i.e., AP, SP, and k-medoids). In this study, the number of clusters (k) is 3, 4, and 5.

Choosing the best clustering results by computing cluster validity indices

In this step, we choose silhouette index as validity indices for selecting the optimal clustering model.

The silhouette index (SI), proposed by Peter J. Rousseeuw in 1986, is an evaluation metric for clustering algorithms. It combines the separation and cohesion of clustering[42]. It is defined as follows:

$$SI\left(X_{\check{i}}\right) = \frac{b_avg\left(X_{\check{i}}\right) - a_avg\left(X_{\check{i}}\right)}{\max\left\{a_avg\left(X_{\check{i}}\right), b_avg\left(X_{\check{i}}\right)\right\}} \tag{23}$$

The value range of $SI\left(X_{\check{i}}\right)$ is $[-1, 1]$. $b_\mathrm{avg}\left(X_{\check{i}}\right)$ represents the average distance between data point $\check{i}$ and its closest cluster, and $a_\mathrm{avg}\left(X_{\check{i}}\right)$ represents the average distance between data point $\check{i}$ and the other data point in the same cluster.

In Table 2, we summarizes the SI index for the clustering approaches in PD1, PD2 and PD3. For PD1, the highest SI value is obtained by applying spectral clustering with (DTW-D + SBD) metric and setting $k = 4$. For PD2, the best SI value is obtained by using hierarchical clustering with (DTW-D + CID) metric and setting $k = 4$. For PD3, the best SI value is obtained by using hierarchical clustering with (DTW-D + SBD) metric and setting $k = 5$.

Analysis and comparisons of the DCSLM

In this section, we validate the performance of the LRFMS segmentation model and discuss the effectiveness of DCSLM. We also discuss the limitations of the study.

Analysis of the obtained clusters

According to the clustering results and the different characteristics of customers, we divide customers into important retention customers, development customers, loyal customers, and general customers in PD1 and PD2[43]. And we divide customers into important retention customers, secondary important retention customers, development customers, loyal customers, and general customers in the PD3.

	PD1			PD2			PD3		
Clustering approach	***k*= 3**	***k*= 4**	***k*= 5**	***k*= 3**	***k*= 4**	***k*= 5**	***k*= 3**	***k*= 4**	***k*= 5**
S+DTW-D	0.4627	0.3759	0.4191	0.7051	0.7088	0.6609	0.5301	0.5525	0.62
S+(DTW-D+CID)	0.4914	0.4022	0.5514	0.7653	0.7615	0.7047	0.8227	0.5558	0.6281
S+(DTW-D+SBD)	0.7112	**0.7506**	0.699	0.6833	0.724	0.7107	0.8441	0.8411	0.5857
H+DTW-D	0.7127	0.7016	0.6664	0.9164	0.9284	0.9151	0.8281	0.8187	0.8495
H+(DTW-D+CID)	0.64	0.5753	0.5229	0.9303	**0.9313**	0.9227	0.8249	0.8143	0.8491
H+(DTW-D+SBD)	0.7005	0.7106	0.6795	0.9096	0.923	0.9058	0.8431	0.8421	**0.8804**
K+DTW-D	0.6812	0.6825	0.66	0.9175	0.9184	0.9101	0.8281	0.8215	0.8479
K+(DTW-D+CID)	0.5403	0.5456	0.5407	0.9201	0.921	0.9126	0.8249	0.8175	0.8474
K+(DTW-D+SBD)	0.6957	0.689	0.6703	0.9104	0.913	0.9038	0.8453	0.8428	0.8666

Table 2. The silhouette index of clustering experiments applied to parts agents (S: Spectral clustering, H: Hierarchical clustering, K: k-medoids). Significant values are in bold.

In order to analyze the resulting customer segments, each customer group needs to be tagged and its behavior analyzed. For analyzing customer behavior, we chose the monetary variable as a representation of customer behavior since the ultimate goal of any firm is to gain the desired profitability[5]. Figures 2, 3 and 4 illustrates each cluster's representative series of customer segments and trend lines obtained using the moving average. The representative series for each cluster is the centroid of that cluster. By analyzing the monetary value of each cluster and corresponding trend lines, we can provide marketing suggestions for each group.

The clustering results on the PD1 are shown in Fig. 2. Cluster C1 exhibits a slight downward trend and is considered a loyal customer group based on its average value. Marketing programs should be designed to prevent the value of customers within C1 from declining. Cluster C2 shows an increasing trend and is considered a development customer group considering the average of the time series values. Effective marketing programs are needed to turn this group into an important customer retention group. Cluster C3 has a stable trend; however, regarding the average value of the series, it is a general customer group. Effective marketing programs to increase the value of customers within C3 should be devised. Cluster C4 shows a downward trend at the beginning of the period, then switches to an upward trend. The cluster is considered an important customer retention group considering the average of the series values. The parts agent should devise effective marketing programs to maintain the profitability of this group.

The clustering results on the PD2 are shown in Fig. 3. Cluster C1 shows a downward trend first and then an upward trend. It is considered a general customer group with a low mean series value. The parts agent should devise effective marketing programs to improve the value of this group of customers. The trend of cluster C2 is growing at the beginning of the period, then declines and becomes steady. C2 is a loyal customer group based on the average value of the series. Specific marketing programs should be designed and executed to prevent customers of the group from becoming general customers. Cluster C3 is a developmental customer group with a downward trend. Marketing programs should be designed to prevent customers of C3 from becoming loyal customers or general customers. In addition, Cluster C4, which first has a downward trend and then an upward trend, is an important customer group with a high mean series value. The parts agent should devise marketing programs to maintain the profitability of this group.

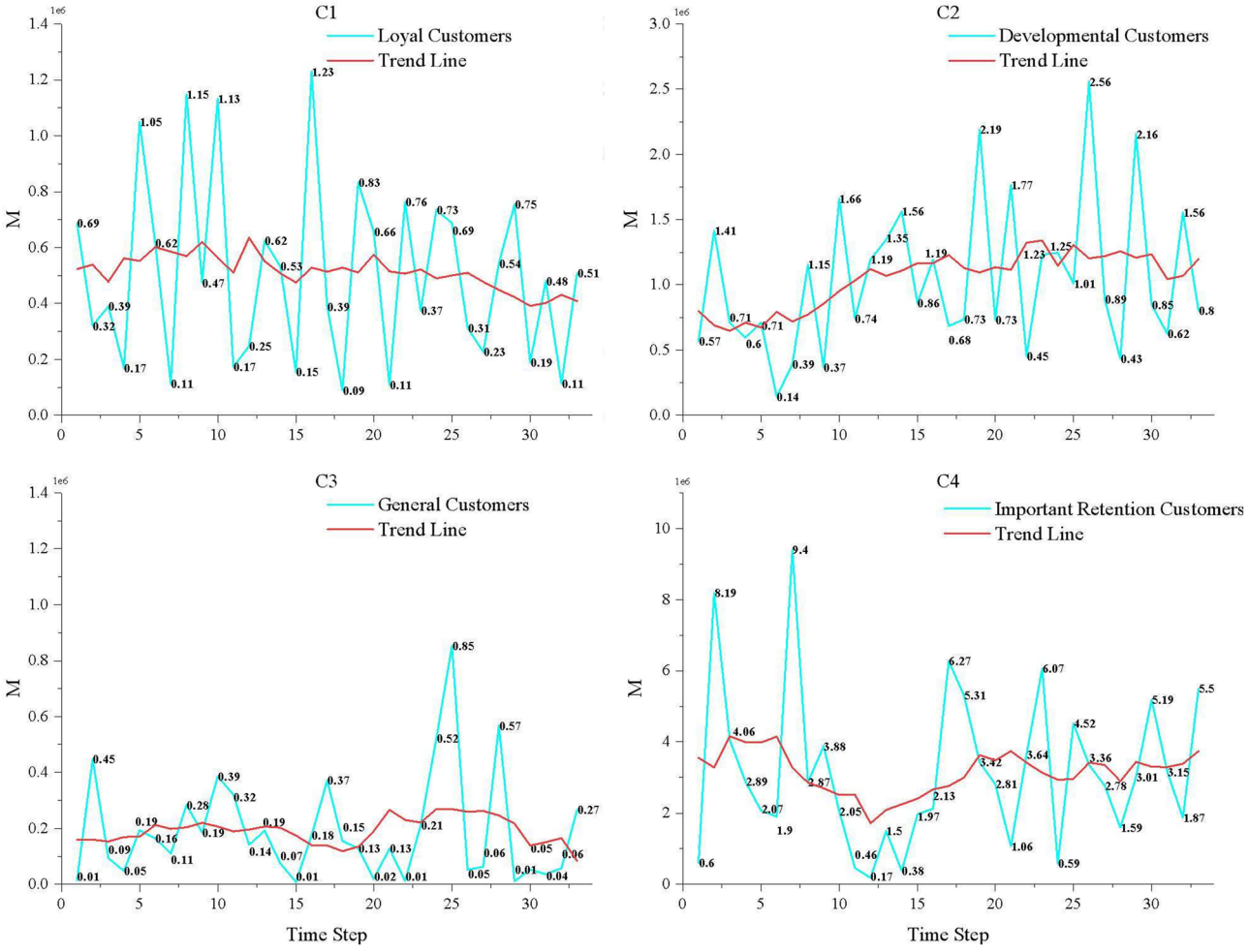


Figure 2. The representative series of each cluster and the corresponding trend line of the PD2.

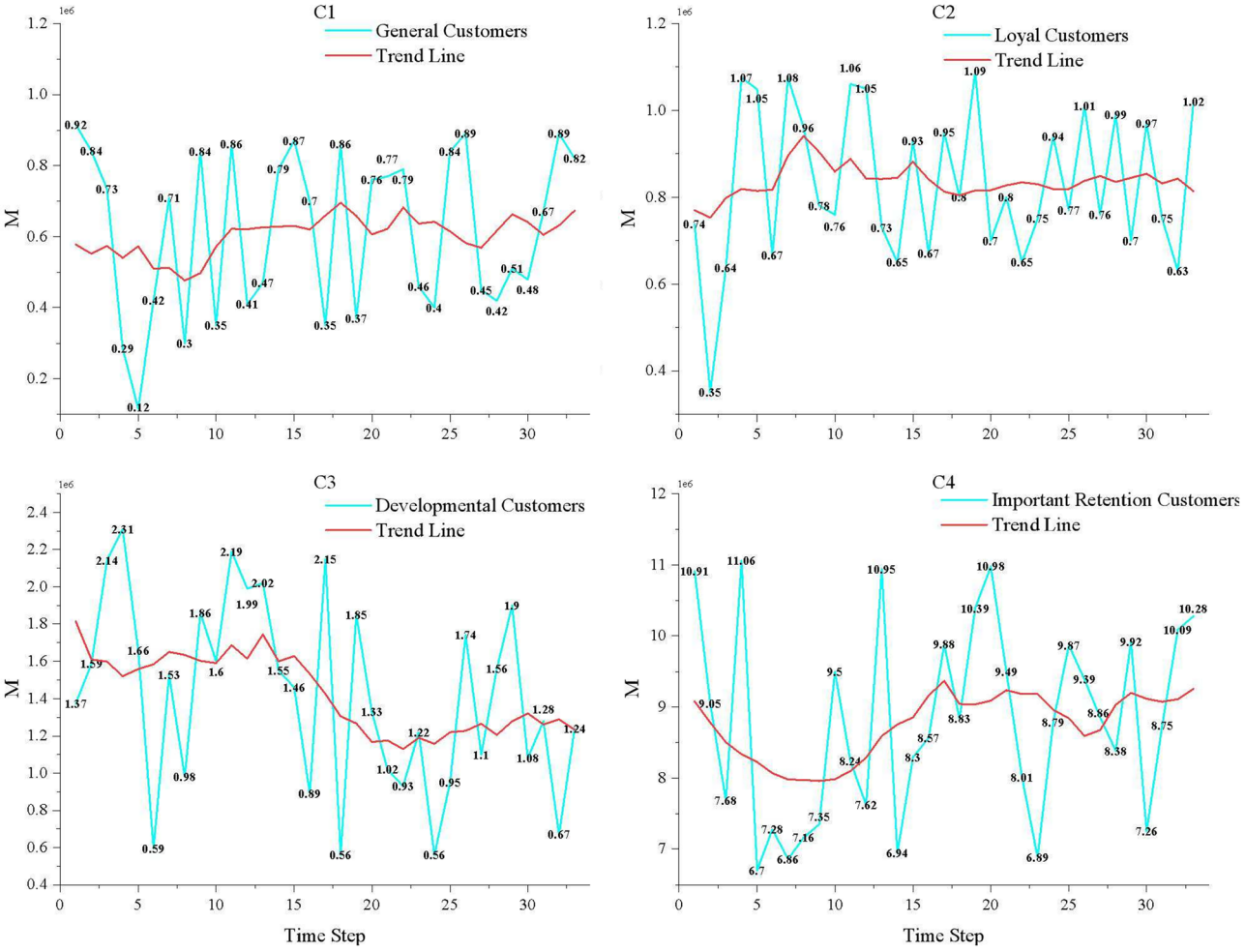


Figure 3. The representative series of each cluster and the corresponding trend line of the PD2.

In PD3, as shown in Fig. 4, the trend of cluster C1 is growing at the beginning of the period, then declines. However, it is a general customer group in terms of its average value. Cluster C2 has a downward trend and is the second important customer group based on the average value of the series. Cluster C3 is a loyal customer group with a downward trend. Cluster C4, which has also undergone a downward trend, is an important customer group with a high mean series value. In addition, cluster C5 is a development customer group whose overall trend is downward. For the five clusters, as they have a downward trend, it is necessary to conduct a thorough investigation to identify the reasons.

Analysis of the LRFMS model

To evaluate the validity of the LRFMS model, we present the segmentation results using the original RFM model and LRFMS model in PD2 to illustrate the credibility of the LRFMS model. The partial results are shown in Tables 3 and 4.

From Tables 3 and 4, we can see that customer with the ID of '1330808365760253952' and '1384039727912718388' have the same classification results under the two models. From Table 3, the customers of '1330808365760253952' and '1384039727912718388' are important retention customers. From Table 4, the two customers have considerable data on the total amount and purchases frequency within 10 days. And the customer of '1330808365760253952' has a better S-value than the customer of '1384039727912718388'. This indicates that the customer with ID '1330808365760253952' is more satisfied with the parts agent, or in other words, has a better user experience. The customers with IDs '1330808365760253967' and '1343940521835801811' are considered to be important retention and developmental customers in the RFM model. In the LRFMS model, it can be observed that these two customers have a high purchase frequency within four days. The customer with ID '1330808365760253967' has a larger total purchase amount, while the customer with ID '1343940521835801811' has a relatively smaller total purchase amount than that of other important retention customers mentioned in Table 4. The customer with ID '1330808365760253967' is considered a development customer due to their lower satisfaction with the parts agent, while the customer with ID '1343940521835801811' is rated as an important retention customer because of their higher satisfaction. Therefore, the LRFMS model can provide a more detailed

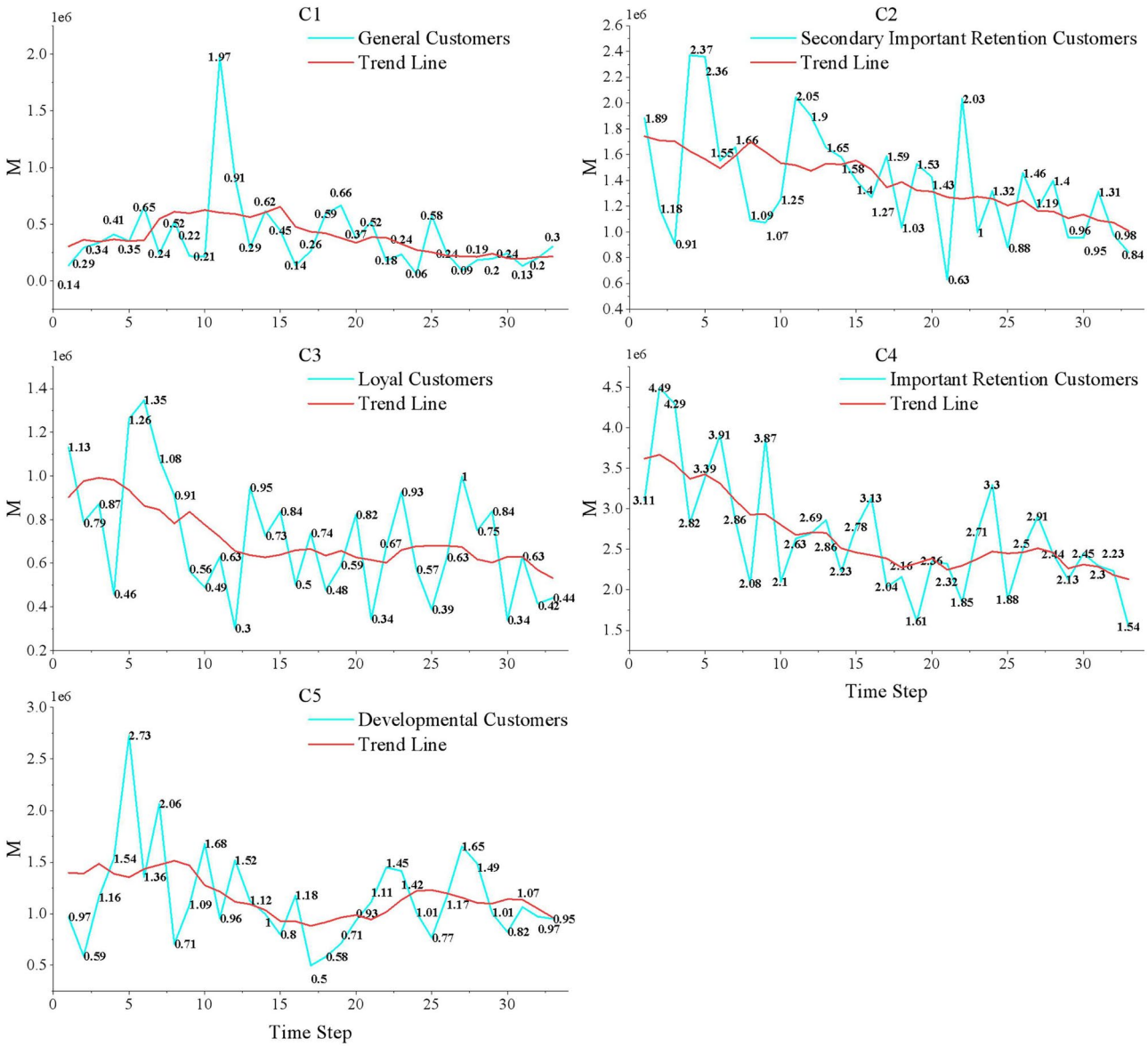


Figure 4. The representative series of each cluster and the corresponding trend line of the PD3.

User ID	R	F	M	Customer segmentation results
1330808365760253952	3	26	555	Important retention customers
1330808365760253967	9	22	725.38	Important retention customers
1343940521835801811	10	17	400.67	Development customers
1364039727912718336	4	3	17	Loyal customers
1364039727912718826	2	4	39. 11	Development customers
...	...	...	...	...
1384039727912718388	4	23	490	Important retention customers

Table 3. RFM model user classification results (partial).

description of user data from the customer's perspective, enhancing the ability to analyze customer value and making the assessment of user value more accurate.

The customer of '1364039727912718336' made a total purchase of parts worth 17.0 RMB, with a frequency of 3 purchases. Since the time of his latest purchase is closer to the end of the purchase period, this customer is classified as a loyal customer in the original RFM model. The customer with ID '1364039727912718826' purchased parts worth 39.11 RMB in total, with a purchase frequency of 4. However, because the time of his last

User ID	L	R′	F	M	S	Customer segmentation results
1330808365760253952	10	4	26	555	8	Important retention customers
1330808365760253967	4	9	22	725.38	4	Development customers
1343940521835801811	4	11.67	17	400.67	10	Important retention customers
1364039727912718336	3	6	3	17	3	General customers
1364039727912718826	5	5.33	4	39. 11	3	General customers
...	...	...	...	...	...	...
1384039727912718388	10	4	23	490	6	Important retention customers

Table 4. LRFMS model user classification results (partial).

purchase is closest to the end of the purchase period, he is classified as a development customer in the original RFM model. According to the analysis of the LRFMS model, the customer with ID '1364039727912718336' made three purchases of low-priced parts within three days and gave a satisfaction rating of 3 to the parts agent. The customer with ID '1364039727912718826' made 4 purchases of low-priced parts within 5 days, with a satisfaction rating of 3 points. Both of these customers have potential commercial value that needs to be further developed, so they are correctly classified as general customers. Therefore, the LRFMS model overcomes the randomness of the R indicator in the original RFM model and more accurately positions the customer.

The experimental results show that the LRFMS model can provide more detailed descriptions of data and overcome the randomness of R indicators in the RFM model. The improved LRFMS model can better achieve one-to-one precision marketing to the customers of the parts agent.

To further evaluate the effectiveness of the LRFMS model, we replaced the LRFMS model in DCSLM with RFM, resulting in DCSRM. We compared the results of DCSLM and DCSRM in PD1, PD2, and PD3, as shown in Table 5. Their performance is evaluated based on three comparison indices: SI, Dunn Index (DI), and Davies-Bouldin Index (DB). The detailed description of DB and DI (including the definition and computational method) has been reported by Saini et al.[44]. Compared to DCSRM, DCSLM showed increases in SI values of 18.7283%, 14.6780% and 21.6190% respectively, and increases in DI values of 35.66%, 28.56% and 33.22% across the three datasets. Furthermore, DCSLM significantly outperformed DCSRM on these datasets in terms of the DB index. These findings indicate that the LRFMS model significantly outperforms the traditional RFM model.

Comparative analysis of different methods

To evaluate the effectiveness of DCSLM, we compare 4 clustering methods, namely clustering method based on DTW and Partitioning Around Medoids (DTW-PAM)[31], Fuzzy Clustering based on Feature Weights (FCFW)[31], clustering method based on Global Alignment Kernel and DTW Barycenter Averaging (GAK-DBA)[45], and Hybrid Clustering for Multivariate Time Series (HCMTS)[46], based on multivariate time series data from LRFMS. The comparison results of the experiments are shown in Table 6. To understand the performance of the DCSLM more clearly, we compare it with several customer segmentation methods, including CATC[13], CBRD[4], TCBS[28], and AFRC[5], as shown in Table 7. To adapt these customer segmentation methods for the MTS dataset, we need to make some modifications based on the literature[31]. From the index values (SI, DB, DI) in Table 6, the mean index values of different clustering methods are calculated, as shown in Fig. 5. The mean index values (SI, DB, DI) for different segmentation methods are computed based on the data in Table 7, as depicted in Fig. 6.

From Table 6 and Fig. 5, we can find that the results of DCSLM can obtain an advantage on three datasets. In addition, one can see that except to DCSLM, the clustering effect of FCFW is also striking. It also obtains superior results in the SI and DI index. In Table 7, DCSLM performs better than other customer segmentation methods overall on three datasets. And it is believed that DCSLM performs significantly better than others in the DI index, but it shows slightly better performance in the other indexes. In Fig. 6, the mean values of DCSLM are overall higher than those of the comparison customer segmentation methods. This means that DCSLM performs better than the other methods on all indices. Therefore, when performing dynamic customer segmentation, DCSLM should be considered to obtain more ideal results.

Dataset	Techniques	Indexes		
		SI	DB	DI
PD1	**DCSLM**	**0.7506**	**0.3802**	**0.2697**
	DCSRM	0.6322	0.4355	0.1988
PD2	**DCSLM**	**0.9313**	**0.0346**	**0.1958**
	DCSRM	0.8121	0.0416	0.1523
PD3	**DCSLM**	**0.8804**	**0.3016**	**0.2747**
	DCSRM	0.7239	0.3903	0.2062

Table 5. Comparison of DCSLM and DCSRM in PD1, PD2, and PD3. Significant values are in bold.

Dataset	Techniques	Indexes		
		SI	*DB*	*DI*
PD1	**DCSLM**	**0.7506**	**0.3802**	**0.2697**
	FCFW	0.6496	1.2216	0.2761
	DTW-PAM	0.661	0.5137	0.1709
	GAK-DBA	0.5726	0.4613	0.1343
	HCMTS	0.565	1.2289	0.1924
PD2	**DCSLM**	**0.9313**	**0.0346**	**0.1958**
	FCFW	0.8887	1.8043	0.181
	DTW-PAM	0.7887	0.5745	0.1802
	GAK-DBA	0.5841	0.497	0.134
	HCMTS	0.6147	1.4415	0.1727
PD3	**DCSLM**	**0.8804**	**0.3016**	**0.2747**
	FCFW	0.6035	2.2072	0.2155
	DTW-PAM	0.5015	0.4199	0.1416
	GAK-DBA	0.5971	0.3993	0.1098
	HCMTS	0.4134	0.7028	0.1419

Table 6. Comparison of different clustering methods in PD1, PD2, and PD3. Significant values are in bold.

Dataset	Indexes	Techniques				
		DCSLM	FCFW	DTW-PAM	GAK-DBA	HCMTS
PD1	***SI***	**0.7506**	0.7419	0.6664	0.6567	0.7419
	DB	**0.3802**	0.7382	0.4637	0.4367	0.7382
	DI	**0.2697**	0.1467	0.0895	0.0772	0.1467
PD2	***SI***	**0.9313**	0.9225	0.9236	0.9079	0.9225
	DB	**0.0346**	0.1187	0.0781	0.0371	0.1187
	DI	**0.1958**	0.0543	0.0523	0.0286	0.0543
PD3	***SI***	**0.8804**	0.8793	0.8653	0.8602	0.8794
	DB	**0.3016**	0.411	0.4712	0.5498	0.9951
	DI	**0.2747**	0.0331	0.0391	0.03	0.0474

Table 7. Comparison of different segmentation methods in PD1, PD2, and PD3. Significant values are in bold.

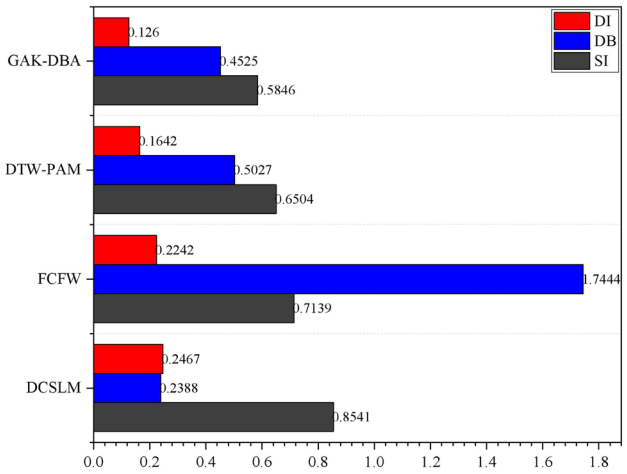


Figure 5. The mean values of different clustering methods under different indices.

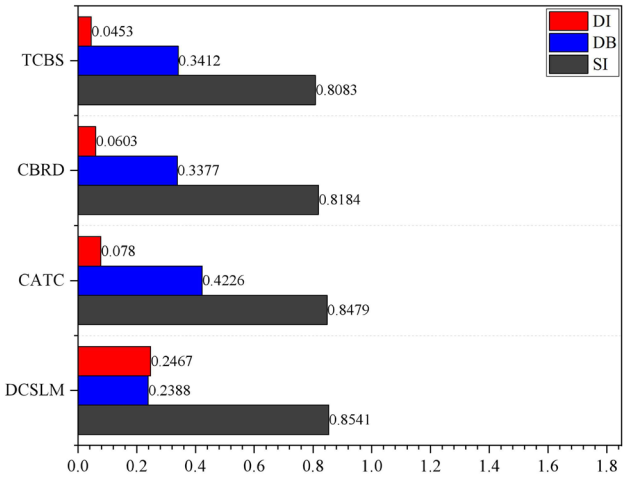


Figure 6. The mean values of different segmentation methods under different indices.

Time complexity

In this section, we analyze the time complexity and time consumption of the comparison algorithms[31]. The time complexity of DCSLM mainly consists of four parts: weight calculation, distance calculation, fuzzy membership matrix construction and result calculation. The time complexity of computing the weights of all the objects is approximately $O(tn)$. The time complexity of processing the distance calculation for all objects is approximately $O\left(n^2t^2 + n^2t\log(t)\right)$. Obtaining the fuzzy membership matrices F_{dtw}, F_{sbd}, and F_{cid} require a time complexity of $O\left(nk^2\right)$ respectively. The complexity of generating the final result is approximately $O\left(n^2I + nkI\right)$, where I represents the number of iterations. The total time complexity is approximately $O\left(\left(I + t^2 + t\log(t)\right)n^2 + \left(k^2 + kI\right)n\right)$. The time complexities of FCFW, GAK-DBA, DTW-PAM, and HCMTS are $O\left(\left(t^2 + t\log(t)\right)n^2 + \left(k^2 + t\right)n\right)$, $O\left(I\left(\left(nt^2\right) + nt\right)\right)$, $O\left(n^2t^2 + kn^2\right)$, and $O\left(n^2kI\right)$ respectively. Similarly, the time complexities of CATC, CBRD, TCBS, and AFRC are also $O\left(n^2t^2 + n^2t + n^2I\right)$, $O\left(n^2t^2\right)$, $O\left(n^2t^2 + nkI\right)$, and $O\left(n^3 + n^2t + n^2I + n^2t\log(t)\right)$ respectively. These time complexities are summarised in Table 8.

Obviously, it is difficult to intuitively see the time efficiency of the algorithm from the time complexity analysis due to existence of the number of objects and iterations. However, the highest order of DCSLM time complexity is the square term, DCSLM requires longer computation time when handling large-scale datasets. We recorded the runtimes of the comparison algorithms on three datasets, as shown in Fig. 7. GAK-DBA has the highest time consumption compared to the other algorithms. This may be because the algorithm requires a large number of iterations to reach convergence. The DCSLM also requires longer calculation time, but its time consumption is within an acceptable range.

Conclusions

In the highly competitive auto parts market, precision marketing and improving customer satisfaction are particularly important. In order to cope with fierce market competition, parts agents need to divide their customers into different groups and gain an in-depth understanding of the characteristics of these groups. To achieve this goal, parts agents use customer segmentation methods to effectively identify distinct groups of customers. RFM models and time series clustering algorithms are essential techniques for dynamic customer segmentation. The original RFM model suffers from the problems of R index randomness and ignoring customers' perceived value.

Techniques	Time complexity
DCSLM	$O\left(\left(I + t^2 + t\log(t)\right)n^2 + \left(k^2 + kI\right)n\right)$
FCFW	$O\left(\left(t^2 + t\log(t)\right)n^2 + \left(k^2 + t\right)n\right)$
DTW-PAM	$O\left(n^2t^2 + kn^2\right)$
GAK-DBA	$O\left(I\left(\left(nt^2\right) + nt\right)\right)$
HCMTS	$O\left(n^2kI\right)$
CATC	$O\left(n^2t^2 + n^2t + n^2I\right)$
CBRD	$O\left(n^2t^2\right)$
TCBS	$O\left(n^2t^2 + nkI\right)$
AFRC	$O\left(n^3 + n^2t + n^2I + n^2t\log(t)\right)$

Table 8. Time complexities of different algorithms.

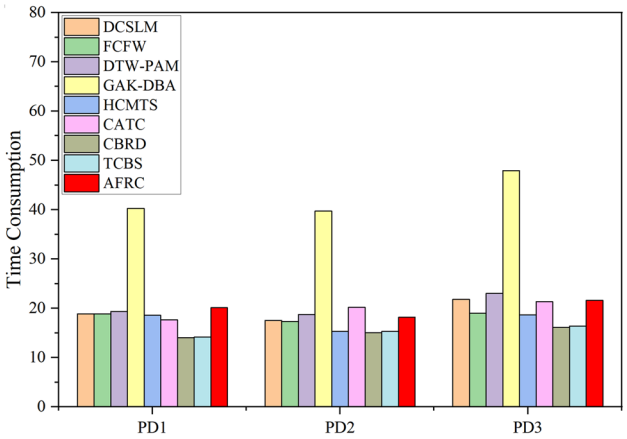


Figure 7. Time consumption comparisons among nine methods.

And for dynamic customer segmentation, the previous research on clustering techniques primarily focuses on univariate time series clustering, with less research on MTS clustering. This paper proposes DCSLM.

DCSLM is improved from two aspects: RFM model and time series clustering. DCSLM represents customer behavior as time series of LRFMS. The LRFMS model is developed by adding the satisfaction (S) variable to the LRFM model, which takes into account the perceived value of the customer and is quite different from previous studies that conducted customer segmentation based on traditional RFM or LRFM models. DCSLM can capture the dynamic behavior of customers more fully by using the LRFMS model. Based on customer time series, MTS clustering is performed to identify groups of customers with similar behavior. The MTS clustering in DSCLM combines state-of-the-art distance measures and some powerful time-series clustering algorithms to obtain the best quality clustering results. The resulting clusters'quality in the MTS clustering task depends on choosing a suitable distance measure and an effective clustering algorithm by computing cluster validity indices[10]. And for dynamic customer segmentation, the previous research on clustering techniques primarily focuses on univariate time series clustering. DCSLM provides a feasible solution for dynamic customer segmentation by using MTS clustering. The effectiveness of the proposed method is verified on real data of auto parts sales.

For the case of this study, real-life data (PD1, PD2 and PD3) from the ASP/SaaS-based manufacturing industry value chain collaboration platform in China is used. It is found that the best clustering result for PD1 is obtained by using spectral clustering with (DTW-D+SBD) metric. And the best segmentation is obtained by applying AP clustering with (DTW-D+CID) metric in the PD2. In addition, for PD3, the best segmentation is obtained by applying AP clustering with (DTW-D+SBD) metric. The customers of parts agents are divided into different customer groups. The segmentation obtained are analysed and behavioral patterns are found. For example, Customers who buy electrical parts (PD3) are divided into five different segments: important retention customers, secondary important retention customers, development customers, loyal customers, and general customers. As the segments obtained for customers all show a downward behavior trend, this parts agent associated with PD3 should conduct a thorough investigation to identify the reasons. On this basis, the management of parts agents flexibly adjusts customer relationship management strategies and marketing methods, and effectively allocates and utilizes resources. This can not only improve customer satisfaction and loyalty, but also maximise the parts agent's profit and maintain a competitive advantage. Furthermore, we conducted extensive experiments on PD1, PD2 and PD3. The experimental results show that the algorithm outperforms other algorithms, further validating the effectiveness of the proposed method.

This study can contribute to prior literature by proposing a dynamic customer segmentation approach based on LRFMS and MTS clustering for the auto parts sales industry. On the other hand, this study proposes important practical directions and a wide range of managerial implications including capturing the dynamic behavior of customers and assisting part agents in developing marketing strategies. Moreover, we have successfully applied DCSLM in the context of the auto parts sales industry, and verified its effectiveness on the third-party platform. However, with the continuous development of the platform and the increasing amount of customer data, DCSLM will gradually expose the problem of insufficient response speed. Due to the huge amount of customer data, DCSLM's segmentation process will become slow and unable to reflect the latest changes and needs of customer groups in time. This is the first limitation. Another limitation of this study is that we only use the transaction data of parts agents' customers, and other customer information, such as geographic location, reputation or browsing history, is not available. In view of this, we plan to introduce an adaptive selection technique into the proposed method in the future. We can use this technology to select the best clustering algorithm in a large dataset to optimise the customer segmentation process and improve time efficiency. In addition, we plan to apply the proposed method to car sales or other fields where more comprehensive customer information can be obtained. Based on this customer information, we will further enhance DCSLM by adding new attributes related to the customer's behavior, and validate the applicability of the method in different areas.

Data availibility

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions
All authors contributed to the study conception and design. Data Curation were performed by Y.Y. L.S. performed the formal analysis. Validation was performed by S.W.. The first draft of the manuscript was written by S.W. and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Competing interests
The authors declare no competing interests.

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